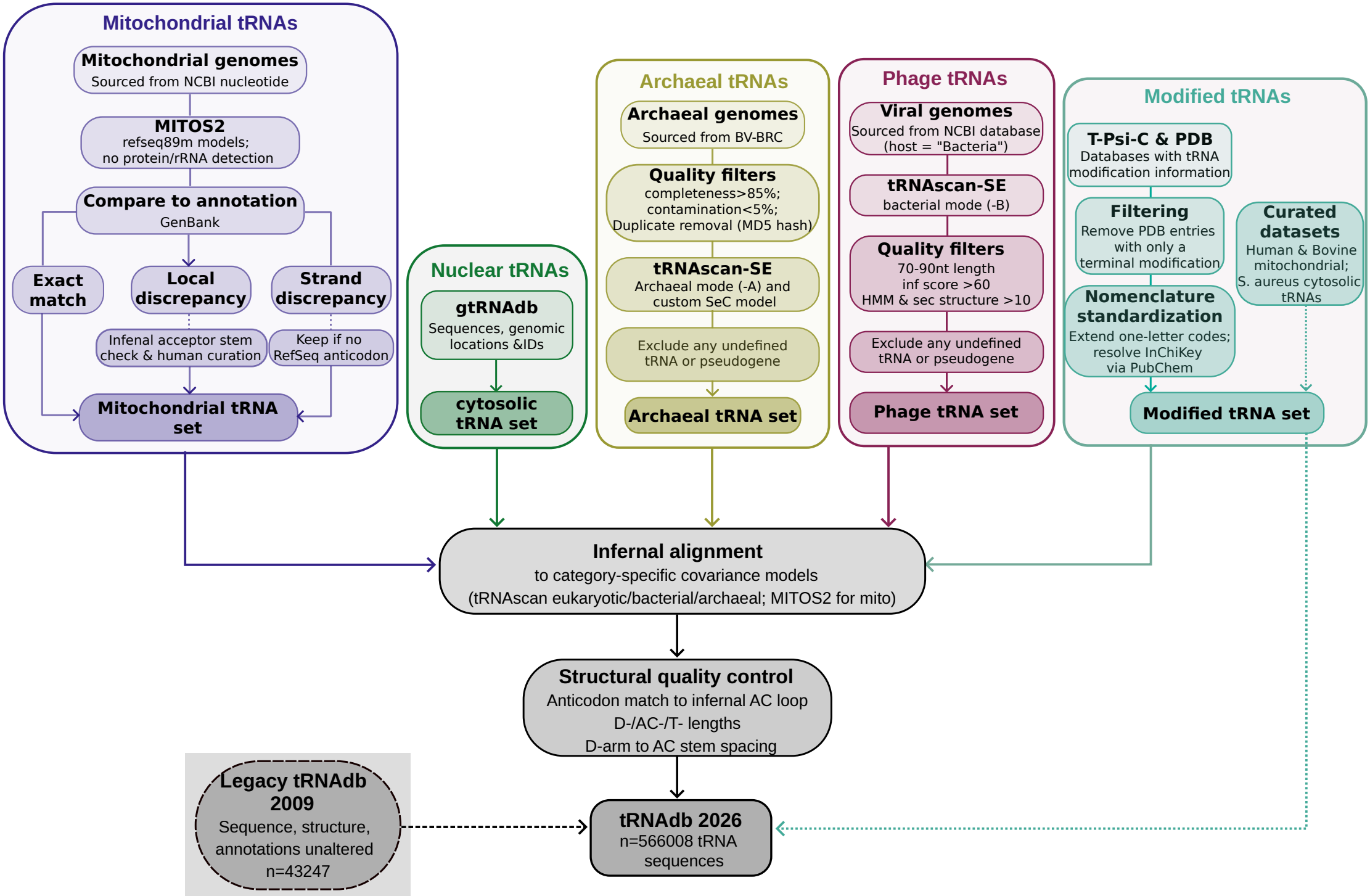
Mitochondrial tRNAs
Mitochondrial genomes
Sourced from NCBI nucleotide
MITOS2
refseq89m models;
no protein/rRNA detection
Compare to annotation
GenBank
Exact match
Local discrepancy
Strand discrepancy
Infenal acceptor stem check & human curation
Keep if no RefSeq anticodon
Mitochondrial tRNA set
Nuclear tRNAs
gtRNAdb
Sequences, genomic locations &IDs
cytosolic tRNA set
Archaeal tRNAs
Archaeal genomes
Sourced from BV-BRC
Quality filters
completeness>85%;
contamination<5%;
Duplicate removal (MD5 hash)
tRNAscan-SE
Archaeal mode (-A) and custom SeC model
Exclude any undefined tRNA or pseudogene
Archaeal tRNA set
Phage tRNAs
Viral genomes
Sourced from NCBI database (host = "Bacteria")
tRNAscan-SE
bacterial mode (-B)
Quality filters
70-90nt length
inf score >60
HMM & sec structure >10
Exclude any undefined tRNA or pseudogene
Phage tRNA set
Modified tRNAs
T-Psi-C & PDB
Databases with tRNA modification information
Filtering
Remove PDB entries with only a terminal modification
Curated datasets
Human & Bovine mitochondrial;
S. aureus cytosolic tRNAs
Nomenclature standardization
Extend one-letter codes;
resolve InChiKey via PubChem
Modified tRNA set
Infernal alignment
to category-specific covariance models
(tRNAscan eukaryotic/bacterial/archaeal; MITOS2 for mito)
Structural quality control
Anticodon match to infernal AC loop
D-/AC-/T- lengths
D-arm to AC stem spacing
Legacy tRNAdb 2009
Sequence, structure, annotations unaltered
n=43247
tRNAdb 2026
n=566008 tRNA sequences